Games in the net

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Abstract. A web site of quiz games has been developed, which incorporates some advances in comparison with most of the known web sites of the same kind. Registered users can play in two different types of single games, in one of them the system allows the player to select the topic of the question. There is also a group game option which deals with several topics. The way in which the system treats each player depends on his level, which is dinamically assigned to him according to his scores. There are also many other posibilities such as record management, rankings, single games management (saving, continuing and deleting games) implemented in the system. It must be emphasized the warnings and expulsions philosofy which punishes those players booked in a group game who do not play propperly .

1 Introduction

1.1 What is this all about?

As we have pointed out previously, a web site of quiz games has been set up.

It not only includes single games and multiplayer games, but also many new aspects, such as the historical data of winners in multiplayer games, management of the different kinds of records (among the great range of forms of the game), ranking of the best players, the current position of each player in this ranking, and so on.

However, there are other posibilities which are much more interesting than the previous ones. For instance, each player can manage their own single games in many different ways: save, continue, delete, ... in this way the system updates the data for the continued games (ie: if you lose a continued game, the game saved previously

will be deleted automatically by the system, therefore you will not be able to continue saved games indefinitely in order to reach better scores and break records).

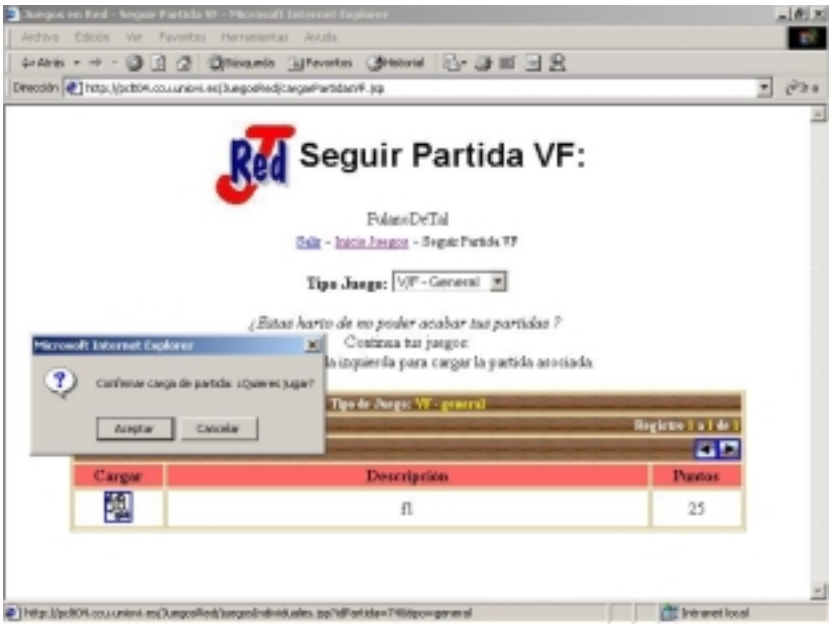


Fig. 1. Dynamic pages that allows players to continue saved games

Furthermore the players are allowed to manage their own questions. As a consequence of that, the players can add, delete, edit and look at the questions that were added to the system by them in the various forms of the game.

The most important point of this web site is its philosophy, trying to make things easier for first timer players and rewarding the best ones. Moreover the possibilities for both of them to win are the same when they play against each other in multiplayer games.

As a result of this idea, a proven level of rank (novice, intermediate, expert) has been added to the recorded data of each player, making this level a great influence over some of the most important parameters throughtout the game.

This level will be updated dynamically. This action will be done according to the scores the player has got in each game. Therefore, the parameters which will be affected by the proccess are: maximum time to answer the questions, number of lifes, score, and so on and so forth.

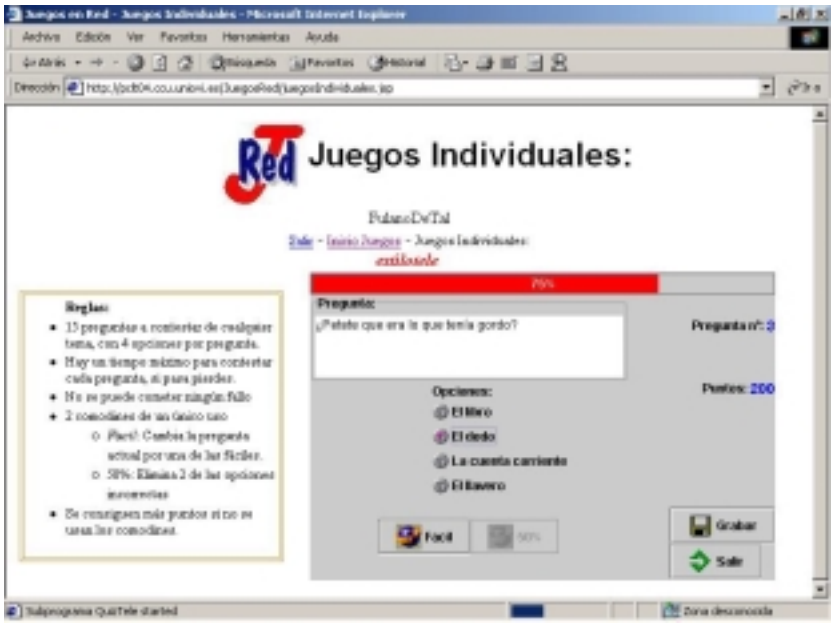


Fig. 2. Single game's aspect (StyleTele mode)

Another interesting point of the system's phylosophy is the way in which the ranking of players is set up: the best players will be rewarded rather than those players who play more.

The result of this is that the players are not classified according to their direct scores. They are given a score based on the game wins they have obtained or the records they have reached.

The multiplayer games are held in accordance with the proposals decided by the players themselves. Each player can step forward to play in a selected subject of the game, in such a way that there will be a period of the time in which any other player can register for the game. After that period of time, the number of players to play will be checked. There must be at least 2 players to play the game, otherwise there will be no game and the registered player will be informed of this situation.

Fig. 3. Page of proposals of games and scores

Multiplayer games are very similar to single games; a collection of 50 questions will be selected at random according to the chosen subject (these questions would be the same for each player). Each player plays according to parameters linked with his/her own level of game. Finally the winner will be the one who takes the least time to get to a higher number of question.

The last point of the philosophy of the game is the one in relation to warnings and expulsions.

Players will get a warning if they don't answer a specific number of questions. In this case, the game will automatically stop and the number of warnings of the player will be updated.

The expulsions will take place whenever a player reaches a maximum number of warnings. At this stage the player will be thrown out of the web site, losing all his games, questions and in general all the information related to this player.

From a potential administrator's point of view (or responsible for the web site), these new guidelines have been incorporated:

1. The configuration of the system: Many of the parameters referred to that are easily configured through a properties file in the server. Some of these parameters are:

- Time to answer (depending on the level).
- Number of lifes (depending on the level).
- Maximum number of players for a multiplayer game
- Maximum number of accumulated warnings before an expulsion
- ...

2. Error controler: Due to both network problems and difficulties accessing to a remote database in each a system, some errors could take place in execution time.

These errors have to be taken into account in order to get an easier resolution and control in the future. This is the reason why an error controler has been set up in order to show, both in the java console of the server and in a *.log file, whatever error happened during the games. The registration of the error includes its description as well as the time and date when it took place.

2 State of the art of the games in The Internet

The network that originally had military and scientific purposes has ended in a kind of big shopping centre in which everything is offered, especially leisure.

In this leisure offer, there are many web sites of any kind of games, from on-line versions of popular comercial games (Quake, Age of Empires, ..) to web sites of registered users, and other games such as puzzles or suchlike.

The reason why commercial games, which are more advanced technologically, are not so required around the The Internet than others is quite simple: they are very slow when they work on-line. This problem is inherent to the net and it is even worse when you realize that the whole game is as slow as the worst reachable speed that the implied players can get. For this reason , web sites of "mini-games" are so important.

Searching in The Internet for similar systems that give ideas and techniques to develop our project, two big groups of systems of quiz games were found:

In the first group, we found games where the player is not registered, so that there is no control over records, saved games...

The second group is made up with games where players are registered and, in general, the best absolute scores are kept (and occasionally the personal scores are kept as well).

The saving of the games of every user is not shown and the only way to propose new questions is through an e-mail address where these questions are accepted or not after being checked.

Games that offer this kind of sites are usually implemented through applets[10] (generally with graphic interface AWT instead of Swing).

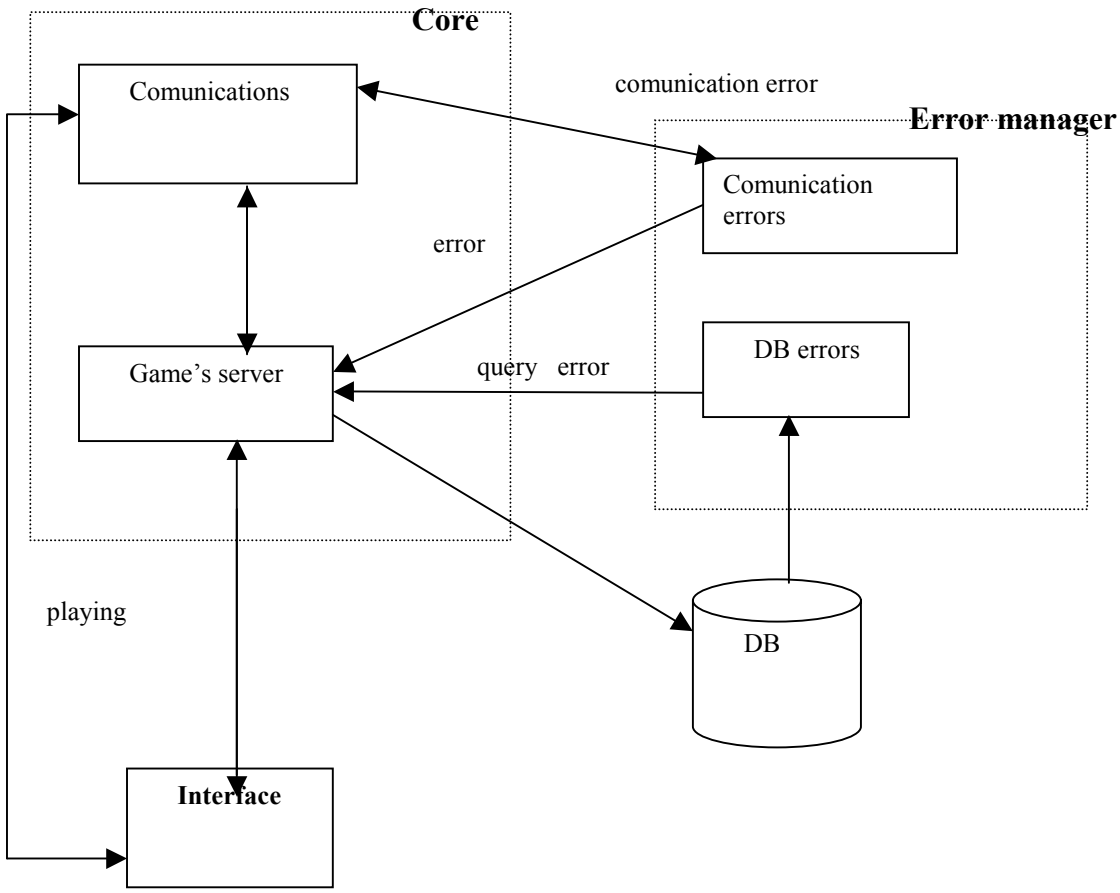
But another possibility was found which is worth a comment. It is about the combination of javascript, css and dynamic pages with forms. The result you get is visually quite attractive and the advantage of this technology (in comparison to applets) is that the client does not need to have installed the last version of JVM in the computer. On the other hand, there are disadvantages due to being simple web pages. For instance: with each answer a different page has to be downloaded (as if you clicked on a link), if there are images they could not come up properly...

An example of this last technical possibility is the web site www.atrivia.com, whereas we would refer to www.cosmoquiz.com or www.trivialnet.com as example of games created with applets.

3 Development

3.1 Architecture

The following diagram will be used to explain the architecture of the implemented system:



1. **Core**: As it is suggested by its own name it is the main block of the system and it has 2 different parts:

   – Comunications: This part is the responsible for the interchange of information between players and server during the games. These connections are also checked by the "connections manager" and if there is a failed connection the server acts as if the game had been left.

- Game's server: It includes not only requests for games but also the querys for rankings, records, questions...The former was developed as a server implemented with java and the latter with JSP pages.

2. **Error manager**: We have to take an special care of the errors because our system depends on the way Intenert works. So we can distinguish between 2 different sources of errors:

   - Comunication errors: They occur during the interchange of information client / server and the operations derived from the requests of games. Synchronization problems with threads and pchysical problems of the net can cause this kind of errors.
   - DB errors: They happen during the access to the remote database. Our labour in this case is write down the situation and make the system keeps on working as usual.

3. **DB**: The database block does not need a detailed explanation, we will just say that it had to be a powerful database system because it supports many simultaneous connections. That's why we chose Oracle.

Besides, as it will be explained further, we took advantage of the pl/sql language from Oracle to have compiled operations in its own engine.

4. **Interface**: It is the visual part seen by each client (user) of the web site. That implies not only the applets (games) but also the resulting html and javascript from the requests of the dynamic pages.

The main detail talking about applets is its size, a code with many small classes and resources such as images or sounds makes the download proccess much slower (a user will not wait eternally) and its execution efficiency worse (a browser's virtual machine is not as efficient as a virtual machine which executes the usual applications).

3.2 Used Technologies.

This is a brief list which contains the technologies used during the development of our project:

- Programming language: **JAVA**
- Server part:
  - Database: **Oracle**
  - Dynamic pages: **JSP**
- Comunications: **RMI**
- Client part:
  - applets
  - javascript
  - html

As it has already been said before we chose Oracle[2] as database system; that was not only because of its power but also because it lets us have compiled procedures and functions (**pl/sql** language) for those frecuent operations which would have been much slower if they had been developed as usual querys or updates(ie: updating records.).

Besides we have taken some advantage of the **DBMS_JOBS** packet (which makes possible to programme the database so it calls a procedure or function with a certain frecuency) to complete the users deleting proccess. When you have to erase a user from the system it is possible that you can not do it straight because he has an absolute record, that is why there is a deleting flag for each user so that every night a database function checks all the player before erasing. Those who have that flag activated and no absolute record at all are the ones that will be erased.

We selected **JSP [11]** instead of **asp**, **php** because it is java and it is very easy to create a comunication with the server by using **javabeans** which were RMI clients.

We developed our comunications with **RMI** because it is very easy to use and above all because we did not need to have anything installed previosly in the client computer. That is why our games are applets, which are RMI clients.

To make this works we just had to include the stub of the game's server in the *.jar file that is downloaded with all the resources of the applet.

Talking about applets we just had to take into account that they are developed with **Swing** and this obliges the client's browser to have a JVM(Java Virtual Machine) which supports JAVA 2 [12][3][13][8][1].

The more extended versions (not the newest) of the more popular browsers do not have this JVM so we had to include the applet with a **object** tag which specifies where to download the proper plugin if it is needed.

As it can be expected **javascript**[4][9] is used to check the format (data's length, null inputs..) of those places where the user complete a form (starting a new session, managing questions..)

This kind of controls are joined (and separated from the html code) into a *.js file so it is easier to adapt it.

3.3 Tests.

Three different types of tests were done:

1. **Code tests**: White box tests and black box tests were done during the implementation period. White box tests check that all the lines in the code are useful and behave the way they are expected (our final code has no death code).To ensure that each part of the final code was working the way is was expected, without take into account the traces, we did black box tests.

2. **Performance tests**: These tests try to check the final behavior of the system under different situations and number of users. The multiplayer (with a maximum of five players) and single games were checked time after time succesfully.

3. **Usability tests**: It will be utterly explained in the following section.

4 Usability tests

4.1 What is usability?

Usability is a metric that measures how well a product can be used by users to reach some specific goals when carrying out a task with effectiveness, efficiency and satisfaction.

By satisfaction we understand the users attitude when using the product (the user may feel comfortable, pleased, have a positive attitude,...) Satisfaction is a subjective factor.

Efficiency can be describe as the speed (with accuracy) in which users can complete the tasks for which they use the product.

Effectiveness is the completeness and accuracy with which users achieve specified goals when carrying out a task.

Efficiency and effectiveness are associated with how easy a system is to learn and to remember, and also with the rate of errors when carrying out the tasks.

4.2 Usability tests

Usability tests were carried out with a sample of 8 subjects who volunteered. These subjects had different levels of experience in the use of the The Internet and data processing in general (4 novice and 4 experts).

The tests assessed 2 different aspects of the system:

1. Visual/formal aspects: We assessed aesthetic of portal, ease of use, adaptation of the texts … It is important to point out that the answers obtained are subjective and because of that some of them might not be valid.
2. Ease of learning: It was verified that the rules of the games and how the system worked were understood. Unlike the questions about visual and formal aspects, the questions were now objective and every question had to be answered. In case the users did not know the answer they had to look for it in the portal. Users were watched by an observer in order to make sure that the process of finding the information was as efficient as possible.

In order for the users to be able to complete the tests they were asked to navigate freely in the system as long as they considered appropriate while testing the portal.

Users were also asked to complete at least the following tasks. They had to:

- create a new player account
- start a game
- manage some questions of their own
- play individual games, saving them and continuing the games several times
- play in groups

4.3 Tests results

Due to the results of the tests some aspects of the portal were changed:

1. Words: Novice users suggested that more colloquial words were used instead of technical terms. For example, the expression "begin session" was changed by "start to play".
2. Help/Rules: Only 2 out of 8 users read the rules of the system before they started to play, as a consequence it took them longer to learn to play well. To solve this problem a summary of the rules was available for the players with every game in a way that is was possible for the player to look at them while the applet was loading or during the game.

The following conclusions can be drawn from the previous tests:

Advantages	Disadvantages
1. Easy to understand 2. Quick learning of how to play the game 3. No player got lost 4. Fast loading of games (less than 10 seconds following Jakob Nielsen´s advice[5][6][7])	1. Complete but poor aesthetics 2. Lack of sound effects 3. Lack of humour and surprise

5 Novelties

We will summarise here some of the aspects that are worth pointing out from all the ones previously explained:

1. Game management: players can manage their own individual games, they can save, delete and continue their games
2. Answers management: similarly to game management, users can manage their own questions and make them part of the game.
3. Dynamic level of players: The users are labelled as novices, intermediate level or experts, according to the scores they get when they play.
4. Dynamic adaptation of the game to the level of the player: Some parameters of the game (response time, scores, number of lifes) depending on the present level of the player, experts are awarded more points while novices are given more time and lives.

5. Warnings and expulsions: In order to promote the efficiency of group games, players are warned when they do not answer a certain number of questions and they are expelled (they are deleted from the BD) when they accumulate a certain number of warnings.

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